Mitigating Evaluative Epistemic Filtering in LLM-Driven Essay Scoring: A Culturally-Aware Multi-Perspective Scoring (CAMPS) Framework

by

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Submitted to the School of Science, Engineering and Technology
in partial fulfilment of the requirements for the degree of

Master of Artificial Intelligence

at the

RMIT UNIVERSITY

September 2026

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Declaration

I certify that except where due acknowledgement has been made, this thesis is my own work, that any help received in preparing this thesis has been acknowledged, and that all sources used have been indicated. I certify that the work is original and has not been submitted previously, in whole or in part, for any other academic award. Ethics procedures and guidelines have been followed (RMIT Human Research Ethics exemption, Project 31036). Any use of generative AI tools in the preparation of this thesis has been disclosed in accordance with course policy. Student name: Bao Nguyen

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Acknowledgements

I would like to thank my supervisor, Dr Ginel Dorleon, for his guidance, direction, and foundational work on multi-perspective bias mitigation that inspired the CAMPS framework. I also thank Dr Thuy Nguyen, the course coordinator, for her support throughout the semester, and RMIT University Vietnam and the School of Science, Engineering and Technology for the resources provided during this research.

Summary

A Vietnamese student opens her essay with her grandfather's proverb, "water wears down stone", and argues the way many Asian writing traditions teach: context first, conclusion arrived at rather than announced. A human examiner recognises the skill. An AI grader, trained overwhelmingly on Western text, may see only a missing thesis statement and mark her down, penalising the shape of her reasoning, not her English.

As AI increasingly grades high-stakes essays, this thesis measures that hidden penalty and proposes a fix, CAMPS: a panel of AI judges, each reading the essay through a different cultural lens, with strong disagreement flagging the essay for a human teacher.

The experiments confirmed the penalty. Every AI model tested scored native-English essays above upper-level Southeast Asian essays, the most biased by over a full grade band, hitting intermediate-level writers hardest. The panel removed about a quarter of the penalty on the most biased models, though more judges did not help, and the disagreement signal proved too coarse, so far, to decide alone which essays need a human. No student should be graded by a single cultural lens, and the bias of any AI grader can now be measured before it is trusted.

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Abstract

Large language models (LLMs) increasingly score student essays, yet models trained predominantly on Western text risk treating Anglo-American rhetorical conventions as the universal standard of quality, penalising structurally valid writing from other traditions. This thesis formalises this mechanism as *evaluative epistemic filtering* and develops CAMPS (Culturally-Aware Multi-Perspective Scoring). CAMPS is an inference-time framework in which evaluator agents, each calibrated to a distinct cultural interpretation of the scoring rubric, score each essay independently before consensus aggregation, with a Bias Divergence Score (BDS) quantifying disagreement. Across 200 topic-controlled ICNALE essays (Southeast Asian learners and a native-English cohort) and four model families, every baseline evaluator scored the native cohort above upper-band Southeast Asian writing (0.19–1.18 bands, all 95% confidence intervals excluding zero). A three-agent panel reduced the gap by 23–28% on the two most biased models, left it unchanged where bias was small, and widened it on the least biased; a five-agent extension added no further reduction. Weight sweeps found substantial gap reduction available at negligible accuracy cost, but BDS tracked the disagreement of eighty human raters weakly or not at all (held-out AUC 0.64 and 0.48). Evaluative bias is general and partially correctable at inference time; reliable autonomous detection remains open.

Thesis Supervisor: Dr Ginel Dorleon

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Abbreviations

ASAP	Automated Student Assessment Prize (essay dataset)
AUC	Area Under the ROC Curve
AES	Automated Essay Scoring
AI	Artificial Intelligence
API	Application Programming Interface
BDS	Bias Divergence Score
CAMPS	Culturally-Aware Multi-Perspective Scoring
CEFR	Common European Framework of Reference for Languages
CI	Confidence Interval
CRI	Cultural Rubric Interpretation
ENS	English Native Speaker (ICNALE cohort)
FPR	False Positive Rate
GRA	Global Rating Archives (ICNALE module)
ICNALE	International Corpus Network of Asian Learners of English
IELTS	International English Language Testing System
L1	First (native) language
LLM	Large Language Model
QWK	Quadratic Weighted Kappa
ROC	Receiver Operating Characteristic
SD	Standard Deviation
SEA	Southeast Asia(n)
TOEFL	Test of English as a Foreign Language
WE	Written Essays (ICNALE module)
WEIRD	Western, Educated, Industrialised, Rich, Democratic
WEP	Written Essays Plus (ICNALE module)

Chapter 1

Introduction

1.1 Problem Statement

Automated Essay Scoring (AES) has entered a new era. Where earlier systems counted surface features such as word length, sentence complexity, and spelling, large language models (LLMs) now evaluate what previous systems could not: the quality of a student's argument. Frontier models are increasingly deployed to score high-stakes writing at scale, promising consistent, rapid, and inexpensive assessment. Yet a model that judges *how well a student argues* must hold some standard of what good argumentation looks like, and that standard is not neutral. LLMs derive their evaluative expectations from training corpora that overwhelmingly represent Western, Educated, Industrialised, Rich, and Democratic (WEIRD) cultural contexts; empirical work has shown that widely used models express cultural values closely aligned with English-speaking and Protestant-European countries [Tao et al., 2024].

Consider an illustration. A Vietnamese student opens an argumentative essay with her grandfather's proverb, *nuoc chay da mon* (water wears down stone). She then develops her position the way many Asian academic writing traditions teach: context first, evidence accumulated gradually, the conclusion arrived at rather than announced. A trained human examiner can recognise the coherence of this inductive structure. An LLM evaluator calibrated by its training distribution to expect a thesis statement in the opening paragraph may instead record "lacks a clear thesis" and "argument feels indirect". It may also penalise the hedging forms that signal politeness in her tradition as weak reasoning. If so, she loses marks not because her English is poor, but because the shape of her reasoning is culturally unfamiliar to the model. Figure 1-1 sketches the two shapes.

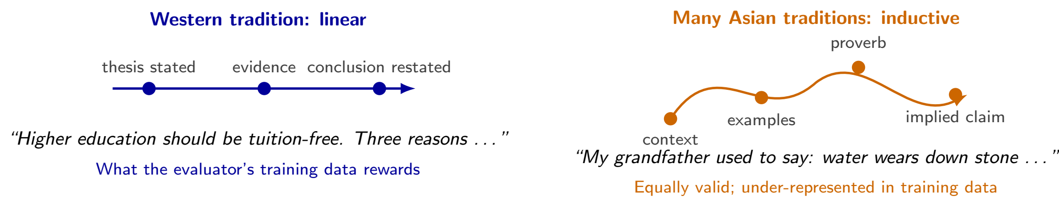

Figure 1-1: Two valid shapes of argument. The linear, thesis-first pattern dominates the text LLM evaluators learn from; the inductive, context-first pattern common in many Asian academic traditions is equally valid but under-represented.

This thesis names this mechanism *evaluative epistemic filtering*: when an LLM evaluates writing, it applies an implicit cultural filter derived from its training distribution, systematically discounting rhetorical strategies that deviate from Anglo-American conventions regardless of their argumentative merit. A pilot study in the preceding phase of this research (Part A) reported statistically significant scoring disparities between Southeast Asian (SEA) and native-English cohorts of equivalent human-rated quality across two frontier models. The same pilot reported that a culturally-diversified scoring panel reduced the disparity. Those findings motivated, but do not yet establish, the claims this thesis investigates at scale.

The problem this thesis addresses can therefore be stated plainly. LLM-based essay scoring applies culturally partial standards, invisibly and at scale, to students whose academic futures depend on the result. The disparity is documented, but no validated framework exists to mitigate it in evaluative settings. Nor does any calibrated mechanism exist to detect the essays it affects. This thesis quantifies the problem across models, builds and tests such a framework, and tests its detection mechanism against human ground truth.

1.2 Research Questions and Hypotheses

Two research questions drive this thesis:

RQ1. To what extent does evaluative epistemic filtering manifest as measurable scoring disparities against Southeast Asian rhetorical patterns in LLM-driven essay scoring, and does it generalise across model families?

RQ2. Can a culturally-aware multi-agent scoring architecture mitigate these disparities at inference time, without retraining or access to model weights, while reliably detecting the essays it cannot fairly score?

These questions are operationalised as four testable hypotheses:

- **H1 (Generalisation).** Evaluative bias against Southeast Asian rhetorical patterns appears as a native−SEA cohort score gap whose 95% confidence interval excludes zero on at least four LLMs, including an open-weight model.
- **H2 (Mitigation).** A five-agent culturally-calibrated panel with a meta-adjudicator reduces the cohort score gap more than the three-agent pilot configuration, without degrading agreement with the corpus's proficiency bands.
- **H3 (Detection).** A Bias Divergence Score threshold calibrated by receiver operating characteristic (ROC) analysis attains at least 90% sensitivity at no more than a 5% false-positive rate against multi-rater human disagreement on the 140-essay Global Rating Archives (GRA) validation set.
- **H4 (Trade-off boundary).** Consensus-weight allocation exhibits an identifiable boundary beyond which cohort-fairness gains degrade scoring accuracy against graded ground truth.

The research questions and hypotheses relate as follows. RQ1 is a question about the *existence and generality* of the problem; it is operationalised by H1, which makes it falsifiable by fixing the scope (four model families) and the test (statistical significance of the cohort scoring gap). RQ2 is a question about the *solution*, and it decomposes into three parts: whether the mitigation works (H2), whether its failures can be detected (H3), and what the mitigation costs (H4). Table 1.1 summarises the mapping and where each hypothesis is tested.

Table 1.1: How the research questions decompose into hypotheses, and where each is tested.

RQ	Hypothesis	Establishes	Tested in
RQ1	H1	Bias exists and generalises across models	Experiment E1
RQ2	H2	The panel mitigates the bias	Experiments E1–E2
RQ2	H3	Residual bias is detectable (BDS)	Experiment E3
RQ2	H4	What mitigation costs in accuracy	Experiment E2

Chapter 5 returns to each hypothesis explicitly. Because several terms recur throughout, Table 1.2 defines them once in plain language.

Table 1.2: Key terms used in this thesis, in plain language.

Term	Meaning in this thesis
Evaluative epistemic filtering	The tendency of an AI grader to mark down writing whose rhetorical form is unfamiliar from its training data, regardless of the argument's quality.
Lens	One AI evaluator given a documented cultural interpretation of the same scoring rubric (Western, Southeast Asian, East Asian, Mekong, or Neutral).
Panel, consensus	Several lenses scoring the same essay independently; their weighted average is the consensus score.
Bias Divergence Score (BDS)	How much the lenses disagree on one essay; high values are meant to flag the essay for a human.
Cohort, cohort gap	A group of essays by writer background (Southeast Asian learners or native-English writers); the gap is the difference in mean score between the two cohorts.
Band	The 1–9 integer score an evaluator assigns; the corpus's own proficiency labels are CEFR bands (B1_2, B2+).

1.3 Theoretical Framing: Epistemic Injustice and Contrastive Rhetoric

Two established bodies of theory ground this thesis. The first is Fricker [2007]'s account of *testimonial injustice*: a speaker is wronged in their capacity as a knower when prejudice in the hearer deflates the credibility of their testimony. Evaluative epistemic filtering can be understood as testimonial injustice automated and scaled: the student's argument is discounted not on its merits but because its form does not match the evaluator's culturally conditioned expectations. Fricker's framework also cautions against a naive remedy:

no evaluator, human or artificial, judges from nowhere. This thesis therefore treats a "culturally neutral" evaluator as a regulatory ideal rather than an achievable design goal, and pursues *plurality* of perspectives rather than pretended neutrality.

The second grounding is contrastive rhetoric. Since Kaplan [1966] first observed that rhetorical organisation is culturally learned, research summarised by Connor [2011] has documented systematic differences between writing traditions. These include the distinction between writer-responsible and reader-responsible coherence, and the validity of inductive, context-first argument structures common in Asian academic writing. This literature supplies the scholarly basis for the cultural rubric interpretations at the centre of the proposed framework, distinguishing them from ad-hoc cultural personas.

The mitigation strategy itself extends the formal multi-perspective bias-reduction framework of Dorleon and Shujaa [2026], who proved that, under ideal conditions, generating text from multiple cultural perspectives with bias filtering can eliminate stereotypical content and perspective omission in *generative* tasks. Companion empirical work documented pervasive cultural bias in LLM treatments of Southeast Asian content [Shujaa et al., 2026]. This thesis asks whether those principles transfer from generation to *evaluation*. Independent support for the core mechanism comes from the evaluation literature: panels of diverse LLM judges have been shown to outperform and de-bias single large judges [Verga et al., 2024], though with diversity defined by model provider rather than by culture. The thesis, in short, tests whether the jury effect holds when the axis of diversity is cultural.

Finally, the fairness–accuracy trade-off at the centre of H4 has a direct lineage in fairness-aware machine learning. Dorleon et al. [2022] formulated feature selection as an explicit trade-off between fairness and prediction performance. FAPFID [Dorleon et al., 2023] then showed that fairness for unprivileged groups can be improved without sacrificing overall performance, through balanced representation. H4 asks whether an analogous frontier exists in evaluative scoring, and where it lies.

1.4 Significance

The significance of this work is threefold. First, equity: as LLM-based assessment becomes a gatekeeper to scholarships, university admission, and migration pathways, a systematic penalty against non-Western rhetorical traditions would constitute discrimination embedded in infrastructure, invisible to its subjects and deniable by its operators. Quantifying that penalty rigorously is a precondition for governing it. Second, method: existing multi-agent AES research improves accuracy without cultural awareness [e.g., Su et al., 2025], while existing debiasing research targets generative outputs rather than evaluative judgements; a validated framework for *evaluative* fairness would fill a documented gap at their intersection. Third, deployability: because the proposed framework operates entirely at inference time through commercial APIs, any institution can apply it without retraining models. This practical property distinguishes it from training-time approaches and matches the constraints of real educational deployments, particularly in the Southeast Asian context where this research is situated.

1.5 Contributions

This thesis makes five contributions:

1. **Conceptual:** it formalises evaluative epistemic filtering as a distinct bias mechanism in LLM-driven scoring, theoretically grounded in epistemic injustice and contrastive rhetoric.

2. **Architectural:** it develops CAMPS (Culturally-Aware Multi-Perspective Scoring), a five-agent culturally-calibrated scoring panel with a meta-adjudicator, extending formal multi-perspective bias reduction from generation to evaluation.

3. **Empirical:** it evaluates the framework on approximately 200 topic-controlled essays from the ICNALE corpus [Ishikawa, 2023] across four frontier and open-weight models, re-establishing the pilot evidence at scale on a corpus purpose-built for the comparison.

4. **Methodological:** it introduces the Bias Divergence Score as a candidate trigger for targeted human review, and evaluates it against multi-rater human ground truth.

5. **Practical:** it characterises the fairness–accuracy trade-off frontier, giving deploying institutions an explicit account of what fairness currently costs.

1.6 Thesis Outline

The thesis follows the research questions in order. Chapter 2 reviews the literature that frames RQ1 and RQ2 and locates the gap this thesis occupies. Chapter 3 develops the CAMPS framework and the method: the data, the three experiments (E1 for H1 and part of H2, E2 for H2 and H4, E3 for H3), the statistical protocol, and the alternatives considered. Chapter 4 reports the experiments in that order and discusses what the results mean, why they arose, how they compare with prior work, and where they are uncertain. Chapter 5 answers each hypothesis, states the contributions and limitations, and outlines future work.

Chapter 2

Literature Review

The literature relevant to this thesis was searched between July and September 2026 using Google Scholar, arXiv, the ACL Anthology, and the Springer and IEEE digital libraries. The search combined the terms “automated essay scoring”, “multi-agent”, “LLM-as-a-judge”, “cultural bias”, “debiasing”, “fairness”, “non-native writers”, and “contrastive rhetoric”. It was supplemented by backward and forward citation chasing from key papers. Every reference cited in this thesis was verified against its primary source. The review is organised thematically: multi-agent and panel-based scoring (Section 2.1), inference-time debiasing (Section 2.2), cultural bias in LLMs (Section 2.3), fairness in AES for non-native writers (Section 2.4), and the contrastive-rhetoric foundations (Section 2.5), before Section 2.6 locates the gap this thesis addresses.

2.1 Multi-Agent and Panel-Based Approaches to Scoring

Two research communities have converged on the same architectural intuition, that several evaluators are better than one, for different reasons.

In automated essay scoring, multi-agent designs pursue *accuracy*. CAFES [Su et al., 2025] orchestrates scorer, feedback-manager, and reflective-reviser agents (average 21% improvement in Quadratic Weighted Kappa, QWK, over direct prompting). Roundtable Essay Scoring [RES; Jang et al., 2025] consolidates trait-based evaluator agents through simulated discussion (up to 34.86% QWK improvement on the ASAP benchmark, zero-shot). Related designs add debate with retrieval [Keramati et al., 2026] and component recognition [Wang et al., 2025]. What unites this family is the axis along which agents differ: *function* or *trait*. None varies the cultural interpretation of the rubric itself, and none evaluates fairness across writer cohorts. Improvements are reported on aggregate agreement. This leaves open the possibility that a more accurate panel is uniformly accurate for some writers and uniformly harsh for others.

In the LLM-evaluation community, panel designs pursue *bias reduction*. Verga et al. [2024] replaced a single large judge model with a Panel of LLM evaluators (PoLL) drawn from three different model families. They found that the panel not only correlated better with human judgement across six datasets but also reduced intra-model self-preference bias, at roughly a seventh of the cost. This is, to date, the clearest empirical demonstration that judge *diversity* counteracts judge *bias*. The diversity axis, however, is model provider, chosen to decorrelate model-specific quirks rather than to represent any substantive difference in evaluative standards. The present thesis can be read as a targeted transplant. It takes the panel-of-judges mechanism whose bias-reducing effect Verga et al. [2024] established, and re-axes the diversity from provider to *cultural rubric interpretation*. Here the bias to be countered is cultural rather than idiosyncratic.

2.2 Inference-Time Debiasing

A second literature attacks bias at inference time, without retraining. BiasFilter [Cheng et al., 2025] periodically evaluates candidate continuations during generation against a fairness reward model and discards low-reward segments, enforcing fairness on *generated text* for both open-source and API-based models. The framework demonstrates that meaningful debiasing is achievable purely at inference time, a design constraint this thesis shares. But its unit of intervention is the output token stream, which has no analogue when the model's output is a single scalar score.

Closest to the present work is Multi-Agent Cultural Debate (MACD) [Tan et al., 2026], which assigns LLM agents explicit cultural personas and orchestrates deliberation. MACD substantially raised the rate of unbiased responses over single-model baselines, and its central finding, that functionally-roled agent frameworks lack the explicit cultural representation cross-cultural fairness needs, independently corroborates the design principle behind CAMPS. The two frameworks nonetheless answer different questions. MACD debiases *what a model says* in open-ended generation. CAMPS debiases *how a model judges*: the deliverable is a score whose fairness can be measured cohort by cohort, and disagreement between cultural agents is exploited as a detection signal rather than dissolved through debate.

A caution for all multi-agent designs comes from the MALIBU benchmark [Mirza et al., 2025], which shows that LLM-based multi-agent systems can implicitly *reinforce* social biases, and that persona-conditioned judging can over-correct. For this thesis, MALIBU motivates two design decisions: consensus weights are calibrated empirically rather than assumed benign (Hypothesis H4 measures the over-correction frontier directly), and panel behaviour is validated against multi-rater human ground truth rather than taken on faith.

A separate line of work mitigates bias inside the model rather than around it. Self-debiasing exploits a model's own ability to recognise undesirable properties of its output, suppressing them at decoding time [Schick et al., 2021], and FairSteer intervenes on hidden-layer activations with dynamically applied steering vectors at inference time [Li et al., 2025]. Both are effective in their settings, but both are white-box methods: they require access to output distributions or internal activations that commercial scoring deployments accessed through an API do not expose. CAMPS was deliberately designed to operate at the level of prompts and outputs alone, trading the precision of internal intervention for applicability to any evaluator, including closed models.

2.3 Cultural Bias in Large Language Models

That LLMs carry a Western-centric evaluative disposition is no longer seriously contested. Tao et al. [2024] compared five OpenAI model generations against nationally representative values-survey data for 107 countries. They found that model responses consistently resemble those of English-speaking and Protestant-European populations. This is the WEIRD alignment that this thesis takes as the distal cause of evaluative epistemic filtering.

The proximal framework for this thesis comes from the supervisor's research programme. Dorleon and Shujaa [2026] formally defined bias categories for generative outputs (stereotyping, omission, ethnocentrism, simplification) and proposed multi-perspective generation with bias detection, proving that under ideal conditions the method eliminates stereotypical content and perspective omission. The companion study [Shujaa et al., 2026] evaluated seven state-of-the-art LLMs on prompts embedding culturally sensitive assumptions about Vietnam, Myanmar, and Nepal, documenting pervasive stereotyping, linguistic ethnocentrism, and epistemic bias, with substantial variation across models. The same group's work on framing-bias detection [Shujaa and Dorleon, 2025] reflects the detection-oriented strand of this programme that the Bias Divergence Score continues: bias treated not only as something to remove but as something to *measure and flag*.

Two further studies shaped how cultural calibration is implemented in this thesis. SEA-HELM, a community-built evaluation suite for Southeast Asian languages, argues that benchmarks produced by machine translation or synthetic generation without community input lose cultural authenticity [Susanto et al., 2025]. The same reasoning led this thesis to ground its cultural lenses in the contrastive rhetoric literature and in a corpus built in the region, rather than in ad hoc persona descriptions. More cautionary still, Khan et al. [2025] showed that survey-based measures of LLM cultural alignment are unstable across presentation formats and that prompting a model to represent a specific culture is not reliably steerable. For CAMPS this is a design constraint, not a refutation: lens prompts are fixed verbatim (Appendix A), decoding is deterministic, and the panel's behaviour is validated empirically against human raters (E3) rather than assumed correct because the prompt requested it.

2.4 Fairness in AES for Non-Native Writers

Concerns about equitable automated scoring of non-native writing predate the LLM era. Vajjala [2018] showed on the TOEFL11 and FCE corpora that the linguistic features most predictive of essay quality differ across L1 populations, evidence that a single scoring model implicitly encodes population-specific assumptions about what good writing looks like. What has changed with LLM-based scoring is the depth at which such assumptions operate:

where feature-based systems encoded them in engineered predictors that could be inspected, LLM evaluators encode them in an opaque prior over rhetorical form [cf. Gallegos et al., 2024].

The LLM-era evidence is accumulating quickly. Pack et al. [2024] evaluated four prominent LLMs (PaLM 2, Claude 2, GPT-3.5, GPT-4) as scorers of English-learner placement essays, finding usable reliability for the strongest model but meaningful variation across models and administrations. Evaluative behaviour is therefore model-specific, which motivates testing H1 across at least four model families rather than generalising from one. Most directly relevant, Gayed [2026] fine-tuned an open-weight model (Gemma-3-27B) for essay scoring and evaluated it on the full TOEFL11 corpus of 12,100 essays across eleven L1 backgrounds, explicitly analysing scoring effects by first language. The study documents L1-linked scoring variation on exactly the corpus family and task that concern this thesis. Like the broader AES fairness literature it extends, however, it stops at *measurement*: no mitigation framework is proposed, and the native/non-native design this thesis requires is out of scope because TOEFL11 contains no native-English cohort. Related recent work has begun to probe mechanism as well as outcome. Yang et al. [2025] showed that prompt-based LLM scorers can infer writer demographics, including English-learner status, from essay text alone. They also showed that scoring behaviour shifts once such a status is recognised.

Taken together, this literature establishes the disparity, its model-dependence, and hints at its mechanism, while leaving the mitigation side of the problem essentially untouched. That asymmetry between diagnosis and treatment is the practical motivation for the framework developed in Chapter 3.

2.5 Contrastive Rhetoric and Cultural Writing Traditions

The claim that rhetorical organisation is culturally learned, and that an evaluator calibrated to one tradition will therefore systematically misread another, has a research lineage of its own. Kaplan [1966] first argued that paragraph organisation reflects culturally specific "thought patterns", contrasting the linear development expected in English academic prose with the patterns he observed in other traditions. The framework was refined substantially in later work: Hinds [1987] distinguished *writer-responsible* languages, in which the burden of clarity falls on the writer through explicit signposting, from *reader-responsible* languages, in which coherence is expected to be inferred by the reader. Under this typology, the "missing thesis statement" an LLM penalises may simply be reader-responsible coherence operating as designed. Connor [2011] consolidated this field as intercultural rhetoric, correcting early essentialist readings: rhetorical traditions are conventions distributed across writing communities, not fixed properties of individuals. This thesis adopts that distinction explicitly, treating L1 cohorts as proxies for rhetorical tradition rather than for identity.

For Southeast Asian learner writing specifically, the empirical base is anchored by the ICNALE project [Ishikawa, 2023], whose topic-controlled design allows rhetorical variation to be observed under matched task conditions across ten Asian countries and a native-English reference cohort. This literature performs two duties in the present thesis. First, it supplies the content of the cultural rubric interpretations in Chapter 3, where each Cultural Rubric Interpretation (CRI) prompt is grounded in documented rhetorical conventions rather than improvised cultural personas. Second, it supplies the defence against the objection that culturally-calibrated agents merely role-play stereotypes. The calibrations are drawn from a half-century of contrastive-rhetoric scholarship, and their behaviour is tested empirically in Chapter 4.

2.6 The Research Gap

Four literatures converge on the problem this thesis addresses, and each stops short of it in a characteristic way. The multi-agent scoring literature (Section 2.1) established that panels of evaluators outperform single judges (CAFES and RES on accuracy, PoLL on bias). But it defines panel diversity by function, trait, or model provider, never by cultural interpretation of the rubric, and it does not measure fairness across writer cohorts. The inference-time debiasing literature (Section 2.2) established that bias can be mitigated without retraining, and MACD specifically established that *explicit cultural representation* in agent frameworks matters. But its object is generated text, not evaluative judgement, and it dissolves inter-agent disagreement rather than exploiting it. The cultural-bias literature (Section 2.3) established the WEIRD disposition of current models and, in Dorleon and Shujaa [2026], provided a formal multi-perspective mitigation framework with provable guarantees, for generation only. Meanwhile, the AES fairness literature (Section 2.4) continues to document L1-linked scoring disparities without offering a validated mitigation framework.

The intersection itself remains empty: culturally-diverse rubric interpretation, applied to evaluative scoring, with inter-agent divergence used as a calibrated trigger for human review and validated against multi-rater human ground truth. To the best of the searches described above, no published work occupies it. That intersection is precisely where CAMPS operates, and the recency of the nearest neighbours on either side suggests the space will not remain unoccupied for long.

Chapter summary

Two literatures converge on the idea that several evaluators are better than one, but neither varies the *cultural* reading of the rubric, and the fairness literature documents L1-linked scoring disparities without a validated remedy. Chapter 3 builds the framework that fills this gap and the experiments that test it.

Chapter 3

Methodology

This chapter sets out what was done and why. It states the working theory (Section 3.1) and specifies the CAMPS framework (Sections 3.2 and 3.3). It then describes the data (Section 3.4) and the three experiments (Section 3.5), records the alternatives considered and the ethical constraints (Sections 3.6 and 3.7), and fixes the metrics and statistical rules before any result is examined (Section 3.8).

3.1 Working Theory: Evaluative Epistemic Filtering

The framework developed in this chapter rests on a working theory of why LLM evaluators systematically under-score culturally unfamiliar writing. This section states that theory precisely, identifies the conditions that sustain the phenomenon, and derives the observable signatures that the experiments in Chapter 4 test.

Definition 3.1 (Evaluative epistemic filtering)**.** *Let E be an essay, $f_{\mathrm{human}}(E)$ the score assigned by a trained human examiner under a rubric R, and $f_{\mathrm{LLM}}(E \mid \theta)$ the score assigned by an LLM evaluator whose parameters θ were estimated from a training distribution $\mathcal{D}$. Evaluative epistemic filtering occurs when*

$$\mathbb{E}[\, f_{\mathrm{LLM}}(E \mid \theta_{\mathcal{D}}) - f_{\mathrm{human}}(E) \,|\, c(E)] \neq 0, \tag{3.1}$$

where $c(E)$ denotes the rhetorical tradition of the essay, and the deviation is systematic in *c: essays whose rhetorical form is under-represented in $\mathcal{D}$ receive systematically deflated scores relative to human judgement, independent of argumentative merit.*

The definition makes two commitments. First, the filter is a property of the *evaluator*, not of the essay: the same essay receives different treatment under evaluators with different training distributions. Second, the deviation is conditional on rhetorical tradition rather than on language proficiency. This is what distinguishes evaluative epistemic filtering from the legitimate scoring of weaker English. It is also what the band-restricted cohort design of Section 3.4 and the human-anchored check of Section 4.4 are built to isolate.

Four conditions jointly sustain the filter. First, training corpora over-represent WEIRD cultural contexts, whose value orientation models demonstrably absorb [Tao et al., 2024]. Second, the standards LLM evaluators apply are opaque, distributed across parameters rather than inspectable as engineered features [Vajjala, 2018, Gallegos et al., 2024]. Third, diagnostic effort has concentrated on generative outputs, leaving the bias categories of Dorleon and Shujaa [2026] without an evaluative counterpart. Fourth, the observable symptom, L1-linked scoring disparity, is documented [Gayed, 2026] but unconnected to any mitigation mechanism.

The theory yields three testable signatures, mapping onto the hypotheses of Chapter 1. The first is a systematic scoring gap between an upper-band learner cohort and the native reference, beyond what residual quality difference can explain (H1). The second is shrinkage of that gap under a panel whose members embody different rubric interpretations, because their individual filters are partially decorrelated (H2). The third is measurable panel disagreement on precisely the essays where filtering operates, making divergence itself a detection signal (H3). The fairness–accuracy boundary (H4) follows from the consensus mechanism below.

3.2 The CAMPS Framework

CAMPS (Culturally-Aware Multi-Perspective Scoring) operationalises a single design principle: if no evaluator judges from nowhere (Section 1.3), fairness is pursued through a *plurality* of explicitly declared perspectives rather than a pretended neutrality. In practice this means five cultural rubric interpretations (Table 3.2), each declared in an inspectable system prompt rather than hidden in model weights. The framework operates entirely at inference time, requires no retraining or weight access, and treats every underlying LLM as a black box reachable through an API. Figure 3-1 shows the pipeline; the five components are specified below.

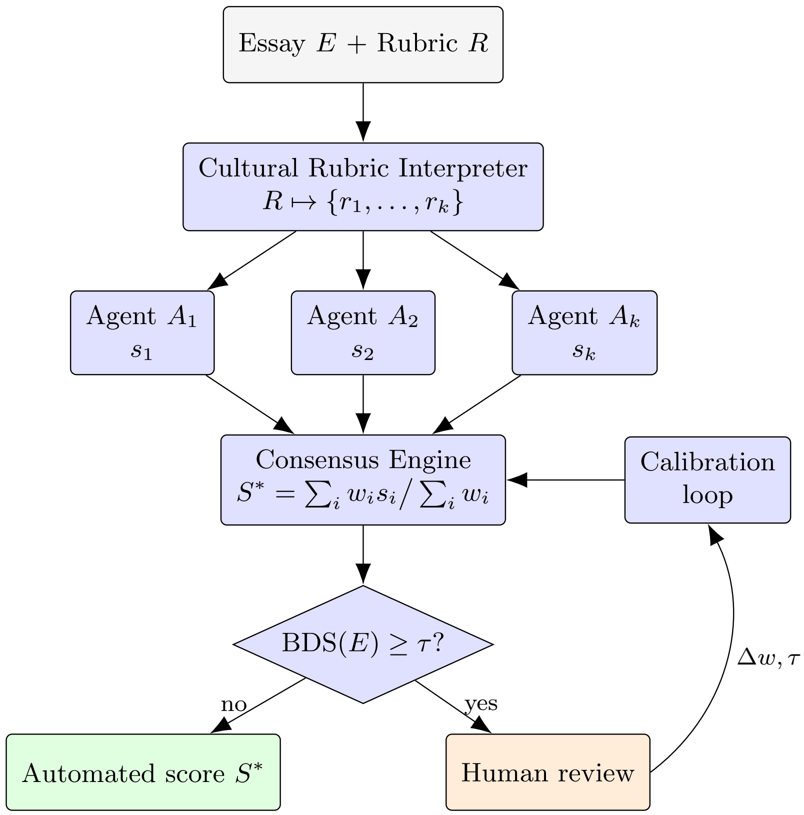

Figure 3-1: The CAMPS pipeline. Each essay is scored in parallel by k *culturally-calibrated* agents: LLM evaluators whose system prompts embed a documented cultural interpretation of the scoring rubric (Component 1). The consensus engine aggregates the scores. The Bias Divergence Score routes contested essays to human review, whose outcomes feed the calibration loop. CAMPS components are shaded blue; inputs and outcomes are unshaded.

Component 1: Cultural Rubric Interpreter (CRI)

Given the scoring rubric R, the CRI produces k culturally-adapted interpretations $\{r_1, \ldots, r_k\}$ through system-level prompting. Each interpretation is grounded in the contrastive-rhetoric literature reviewed in Section 2.5 rather than in improvised personas. The Southeast Asian interpretation, for example, instructs the agent to recognise inductive thesis development and reader-responsible coherence [Hinds, 1987] as valid realisations of the rubric's coherence criterion. The neutral interpretation is retained as a regulatory ideal in Fricker's sense [Fricker, 2007], not as an achievable standpoint. The full prompt texts are reproduced verbatim in Appendix A.

Component 2: Evaluator agent pool

Each interpretation is bound to an independent evaluator agent, and each agent scores the essay without access to the other agents' outputs:

$$s_i = A_i(E, r_i), \qquad i = 1, \ldots, k, \tag{3.2}$$

with temperature fixed at $t = 0.0$ and structured (JSON) output, so that every run is deterministic and reproducible. Independence at this stage is deliberate: the MALIBU results [Mirza et al., 2025] caution that interacting agents can amplify rather than cancel bias, so CAMPS confines interaction to the aggregation stage, where it is mathematically controlled.

Component 3: Consensus aggregation

The panel's scores are combined as a weighted mean,

$$S^*(E) = \frac{\sum_{i=1}^{k} w_i \, s_i(E)}{\sum_{i=1}^{k} w_i}, \qquad w_i \geq 0, \tag{3.3}$$

with weights initialised equal. Because S^* is linear in the agents' scores, the consensus cohort gap is the same weighted combination of the individual lenses' cohort gaps. Reweighting the panel therefore trades the gap against agreement with graded ground truth. This is the mechanism behind the fairness–accuracy boundary hypothesised in H4, and Section 3.5 describes how the weight space is swept to locate it empirically.

Component 4: Bias Divergence Score (BDS)

The framework's detection signal is the dispersion of the panel:

$$\mathrm{BDS}(E) = \sqrt{\frac{1}{k}\sum_{i=1}^{k}\big(s_i(E) - S^*(E)\big)^2}. \tag{3.4}$$

A low BDS means the cultural interpretations converge and the automated score can be trusted. A high BDS means the lenses disagree, which under the working theory is the signature of an essay whose score depends on culturally contingent expectations. Such an essay is flagged for human review. Part B replaces the pilot's hand-set threshold ($\tau = 0.75$) with one calibrated by ROC analysis against multi-rater human ground truth (H3; Section 3.5).

Component 5: Calibration loop

Outcomes of human review feed back into the framework in two ways: the consensus weights w_i are re-optimised to maximise agreement with human scores subject to a cohort-fairness constraint, and the threshold τ is re-estimated as review outcomes accumulate. The loop closes the system without touching model weights; everything learned lives in w and τ. The loop is specified as part of the framework but not exercised in this thesis, whose experiments evaluate a single calibration pass (E2's weight sweep and E3's threshold calibration).

The five components divide the framework's labour: the first two create and apply the plurality of perspectives, the third converts plurality into a single defensible score, and the last two turn residual disagreement into detection and adaptation. No component alone mitigates the filter; the fairness properties of CAMPS are a property of the assembly. Table 3.1 positions CAMPS against the frameworks reviewed in Chapter 2 along the five capabilities the design combines.

Relative to the Part A pilot, the framework specified here differs in three respects, each developed in the following sections. The panel is scaled from three to five cultural interpretations with a meta-adjudicator (Section 3.3). The corpus and cohort design are rebuilt on

Table 3.1: Capability comparison. *Inf.* = inference-time only; *Cult.* = explicit cultural perspectives; *Eval.* = evaluative (scoring) task; *Div.* = divergence-based detection; *HITL* = calibrated human-in-the-loop trigger.

Framework	Inf.	Cult.	Eval.	Div.	HITL
CultureLLM [Li et al., 2024]	×	✓	×	×	×
BiasFilter [Cheng et al., 2025]	✓	×	×	×	×
CAFES [Su et al., 2025]	✓	×	✓	×	×
RES [Jang et al., 2025]	✓	×	✓	×	×
PoLL [Verga et al., 2024]	✓	×	✓	×	×
MACD [Tan et al., 2026]	✓	✓	×	×	×
Dorleon & Shujaa [2026]	✓	✓	×	×	×
CAMPS (this thesis)	✓	✓	✓	✓	✓

data that supports the comparison (Section 3.4). Both the consensus weights and the BDS threshold are calibrated empirically rather than hand-set (Section 3.5).

3.3 Scaling the Panel: from Three to Five Agents

The pilot panel comprised three rubric interpretations: Western, Southeast Asian, and a neutral regulatory ideal. Scaling to five requires a principled answer to *which* lenses to add. A candidate lens must satisfy three criteria. It must be (i) *documented* in the contrastive-rhetoric literature, so the CRI prompt encodes scholarship rather than stereotype; (ii) *validatable* against a corpus cohort from the corresponding tradition; and (iii) *distinct*, expected to diverge from the existing panel on identifiable features. These criteria select an *East Asian* lens, grounded in reader-responsible coherence [Hinds, 1987] and validatable against ICNALE's East Asian cohorts, and a *Mekong/Vietnamese* lens, motivated by the region this research serves and validatable against the WEP cohorts. A Middle Eastern lens fails the validatable criterion for this corpus and is deferred. Table 3.2 summarises the panel.

The second architectural addition is a *meta-adjudicator*: a further LLM instance that receives the five agents' scores and their short written rationales, and produces a reasoned consensus score,

$$S^{\text{adj}}(E) = M\big(s_1, \rho_1, \ldots, s_k, \rho_k\big), \tag{3.5}$$

where ρ_i is agent i's rationale. The adjudicator is blind to writer identity and cohort meta-

Table 3.2: The five-lens panel: rhetorical grounding and the corpus cohort against which each lens's behaviour can be validated.

Lens	Rhetorical grounding	Validating cohort (available)
$\text{CRI}_{\text{Western}}$	Writer-responsible coherence; explicit thesis-first linear development [Kaplan, 1966, Connor, 2011]	ICNALE ENS
CRI_{SEA}	Inductive, context-first development; collectivist framing [Connor, 2011, Ishikawa, 2023]	ICNALE THA, IDN, PHL
$\text{CRI}_{\text{EastAsian}}$	Reader-responsible coherence; delayed thesis; *ki-sho-ten-ketsu*-style organisation [Hinds, 1987]	ICNALE CHN, JPN, KOR, TWN
$\text{CRI}_{\text{Mekong}}$	Proverb- and narrative-anchored argumentation observed in Mekong-region learner writing [Ishikawa, 2023, Shujaa et al., 2026]	ICNALE WEP VNM, KHM, LAO, MMR
$\text{CRI}_{\text{Neutral}}$	Regulatory ideal: argumentative merit irrespective of structural ordering [Fricker, 2007]	(all cohorts)

data and runs at $t = 0.0$. Where the weighted mean treats every judgement as exchangeable evidence, the adjudicator can weigh *arguments*, discounting an agent whose rationale misreads the essay. E2 compares the two aggregation paths directly, with BDS computed from raw agent scores in both conditions. Responding to the caution of Mirza et al. [2025], deliberation is added *after* independent judgement, never during scoring.

3.4 Data

Corpus selection

The design requires a corpus with four properties: Southeast Asian learner essays identified by L1 or country; a native-English reference cohort writing under identical task conditions; a human-assigned proficiency measure; and licence terms permitting research at this scale. A systematic search of the major public learner corpora found exactly one satisfying all four. TOEFL11 [Blanchard et al., 2013] has no Southeast Asian L1 and no native cohort, ICLE v3 lacks Southeast Asian L1s, ELLIPSE lacks L1 labels, and EFCAMDAT lacks a native reference.[1] The ICNALE corpus family [Ishikawa, 2023] satisfies all four and is adopted.

The ICNALE corpus family

ICNALE collects topic-controlled English essays from college students across ten Asian countries and regions, together with a native-English reference cohort writing under identical conditions; the three modules used are summarised in Table 3.3. Critically, every written essay addresses the same two prompts under controlled length and time conditions, removing the prompt-variation confound of TOEFL11's eight-topic design, and all learners carry CEFR-linked proficiency bands (A2–B2+), enabling band-restricted learner-cohort construction.

Table 3.3: ICNALE modules used in this thesis (contents as published on the official ICNALE distribution site, March 2026).

Module	Contents	Role in this thesis
Written Essays (WE) v2.6	5,600 essays, 200–300 words, two controlled topics; ten Asian countries/regions plus native ENS cohort	Primary cohorts: SEA learners and native reference
Written Essays Plus (WEP) v0.7	2,340 essays from additional countries including Vietnam, Cambodia, Laos, Myanmar	Available validating cohort for $\text{CRI}_{\text{Mekong}}$; demo samples (not scored in E1–E3)
Global Rating Archives (GRA) v2.1	140 essays, each rated by 80 raters of varied L1 and occupational backgrounds (11,200 ratings)	Multi-rater human ground truth for BDS validation (E3)

Cohort construction and sampling

Figure 3-2 summarises the sample. The Southeast Asian cohort draws from the Thailand, Indonesia, and Philippines components of WE, restricted to the two upper proficiency bands (B1_2, B2+) and stratified by topic and band. The native cohort draws from ENS, which ICNALE does not band (native writers are the corpus's reference group). Sampling yields

[1] Corpus contents verified against their official distribution pages in July 2026: TOEFL11 via the Linguistic Data Consortium (LDC2014T06), ICLE via UCLouvain, ELLIPSE via its public repository, EFCAMDAT via the EF–Cambridge Open Language Database.

$n \approx 100$ per cohort under a fixed seed. Restricting the learner cohort to its upper bands narrows the proficiency range so that a cohort gap cannot be attributed to weak English, but it cannot equate the cohorts. Any residual quality difference between upper-band learners and native writers remains inside the gap. The gap is therefore interpreted as an *upper bound* on evaluative filtering (Section 3.8), with a human-anchored check on the GRA essays reported in Section 4.4. The GRA's 140 essays are retained in full as E3's validation set.

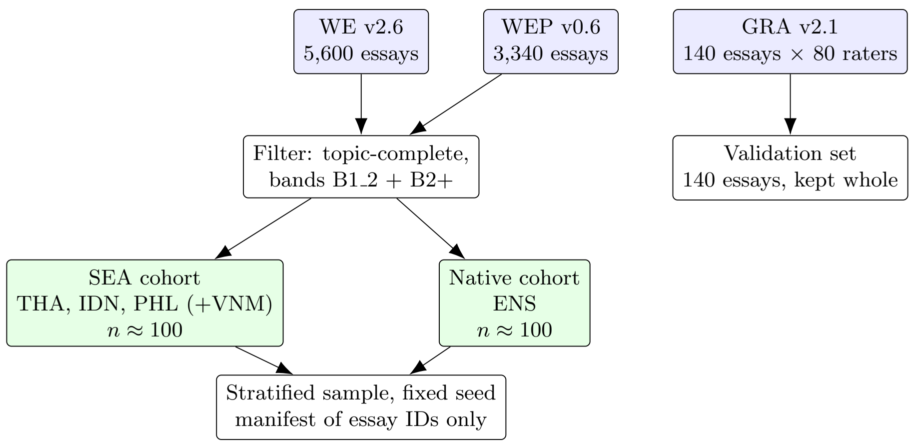

Figure 3-2: Cohort construction and sampling. Essay texts remain within licensed storage; the reproducibility manifest records essay identifiers only.

Scoring target and ground truth

Every agent, including the single-agent baseline, returns an integer band from 1 to 9 on the IELTS Coherence and Cohesion descriptors (the prompts are verbatim in Appendix A). Panel consensus is the mean of these integers and so takes non-integer values. Agreement with ground truth is computed against the corpus's CEFR bands for the learner cohort and against the GRA's multi-rater consensus for the validation set. In the GRA archive each rater scores ten analytic dimensions (0–10) grouped as Language, Content, and Attitude, plus a holistic score out of 100. Per-essay rating dispersion across the 80 raters provides the human-disagreement benchmark against which BDS is compared. Two properties of the archive shape the E3 design. First, all 140 rated essays address the part-time-job prompt, so the validation set is single-topic. Second, the raters disagree substantially among themselves (mean within-essay standard deviation of the Content ratings ≈ 1.8 on the 0–10 scale). Disagreement among evaluators, human or artificial, is the normal condition of essay assessment. The question E3 asks is whether the panel's divergence tracks the essays on which *humans* diverge.

Licence and handling

ICNALE is distributed free for research with redistribution prohibited [Ishikawa, 2023]. Essay texts are stored only in access-controlled storage and never committed to version control; the reproducibility package releases code, prompts, configurations, and essay-ID manifests only. The data are de-identified by the distributor and fall under the ethics exemption of Section 3.7.

3.5 Experimental Design

Three experiments test the four hypotheses (Table 3.4). E1 runs on all four models; E2 and E3 run on the two models with the largest E1 gaps (GPT-4o and Claude Sonnet), confining the costlier extended conditions to the evaluators where mitigation is material. All conditions share the same determinism guarantees: temperature fixed at 0.0, structured JSON outputs, exact model versions pinned in the experiment configuration, and a fixed random seed for all sampling. As a result, every result in Chapter 4 is reproducible from the released configuration, prompts, and essay-identifier manifest.

Table 3.4: Overview of the three experiments: data, conditions, and the hypotheses each tests.

Exp.	Data	Conditions	Tests
E1	SEA + native cohorts ($n \approx 100$ each)	Single-agent baseline vs. CAMPS k=3, four models	H1; H2 (part)
E2	Same cohorts	CAMPS k=5 with and without meta-adjudicator; consensus-weight sweep ($\Delta w = 0.05$); ablation over $k \in \{1, 3, 5\}$	H2; H4
E3	GRA validation set (140 essays × 80 raters)	CAMPS k=5; ROC calibration of τ on a calibration split, evaluation on a hold-out split	H3

Experiment E1 (scaled replication) scores both cohorts under the single-agent baseline and the pilot's three-agent panel on all four models, establishing whether the gap exists at scale (H1) and providing the k=3 reference for E2.

Experiment E2 (extended architecture) scores the same essays under the five-agent panel, with and without the meta-adjudicator. It also sweeps the consensus weights in steps of 0.05 under $\sum_i w_i = 1$, tracing the fairness–accuracy frontier from which H4's boundary is read off empirically. The ablation over k separates the contribution of panel size from that of the adjudicator.

Experiment E3 (detection validation) computes BDS for the 140 GRA essays. Human rating dispersion provides the reference notion of a genuinely contested essay, operationalised as the top quartile of per-essay standard deviation across the 80 raters' Content ratings. The threshold τ is calibrated on a random half of the GRA set (Youden's J) and evaluated on the held-out half. Sensitivity, false-positive rate, and area under the ROC curve are reported.

3.6 Alternatives Considered

The rubric of this thesis requires critical awareness of alternatives; this section records the designs considered and rejected, and why.

Training-time cultural adaptation, as in CultureLLM [Li et al., 2024], fine-tunes models on culturally augmented data. It is rejected on deployment grounds: it requires weight access or fine-tuning budgets unavailable to most educational institutions, must be repeated for every model generation, and cannot be audited by the deploying institution. *Activation steering*, as in FairSteer [Li et al., 2025], intervenes on internal representations and is infeasible for API-only access by construction (Section 2.2).

Single-prompt debiasing, instructing one evaluator to "score fairly across cultures", collapses the plurality of standards back into a single opaque judgement. Bias can then only be asserted away, not measured, and no divergence signal remains to calibrate a review trigger against. The single-agent baseline in E1 provides the nearest empirical comparison, though this specific instruction was not separately run. *Inter-agent debate*, as in MACD [Tan et al., 2026], is rejected because interaction can amplify bias [Mirza et al., 2025] and deliberation before scoring destroys the independence that makes BDS meaningful; CAMPS confines deliberation to the optional post-hoc adjudicator. *Recruiting external human raters* was rejected on ethics-scope and timeline grounds (Section 3.7); the GRA archive supplies eighty raters per essay without new data collection. Alternative corpora were eliminated on design grounds in Section 3.4.

3.7 Ethical Considerations

This research analyses de-identified secondary data only, under an RMIT human research ethics exemption (Project ID 31036). No attempt is made to re-identify any writer, no essay text is redistributed, storage follows the project's data management plan, and no external participants are recruited. Any future extension involving human participants would require a new ethics assessment. Two considerations shape the design rather than merely constrain it. First, cultural essentialism: agents are calibrated to *documented rhetorical traditions*, not nationalities, and L1 cohorts are treated throughout as proxies for rhetorical tradition rather than identity. Second, downstream welfare: CAMPS is human-in-the-loop precisely because the cost of a misclassified essay falls on a student; BDS exists to route uncertain judgements to a human rather than automate them. Generative AI use in this project is disclosed in accordance with course policy.

3.8 Metrics and Statistical Protocol

Primary metric. The primary fairness quantity is the *cohort score gap*: the difference in mean panel score between the native and Southeast Asian cohorts,

$$\Delta = \overline{S^*}_{\text{native}} - \overline{S^*}_{\text{SEA}}, \tag{3.6}$$

computed on an upper-band learner cohort and the unbanded native reference. Absent filtering the gap need not be zero, since native writers may differ in quality from upper-band learners even under a neutral judge. Δ is therefore an upper bound on filtering. The treatment effects of interest are *changes* in Δ between conditions on the same essays, which the residual quality difference does not affect. This operationalisation follows the pilot and is forced by a property of the reference data: the native cohort writes as an unbanded reference group (ICNALE assigns CEFR bands to learners only), so band-agreement statistics are undefined for it. Agreement with graded ground truth is reported secondarily where such truth exists, within the E2 weight-sweep analysis: Quadratic Weighted Kappa,

$$\kappa_w = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}, \qquad w_{ij} = \frac{(i-j)^2}{(R-1)^2}, \tag{3.7}$$

against proficiency bands for the learner cohorts, and against the multi-rater human consensus for the GRA validation set (E3).

Uncertainty. Every reported statistic carries its n. Every cohort gap and gap change is reported with a 95% percentile bootstrap confidence interval (10,000 resamples, fixed seed); band-agreement and detection statistics are reported as point estimates with their n. Following the statistical-reporting guidance adopted for this thesis, confidence intervals are the primary inferential device. p-values are reported only for the pre-registered paired-bootstrap tests of gap change (one-sided probability of no reduction). No post-hoc significance testing is performed on exploratory findings.

Detection metrics. BDS performance in E3 is reported as sensitivity, false-positive rate, and area under the ROC curve on the held-out split, with the calibrated threshold τ stated explicitly.

Weight calibration. The E2 weight sweep reports the full frontier rather than a single optimum. At each weight vector, the cohort score gap (fairness) and κ_w against CEFR bands on the banded learner cohort (accuracy; the native reference is unbanded) are recorded. The H4 boundary is read off the resulting Pareto frontier.

Reproducibility. All tables and figures in Chapter 4 are generated by the analysis scripts in the project repository from the logged scoring calls; no statistic is computed by hand. The configuration file, prompts, essay-identifier manifest, and analysis code are released; essay texts are not (Section 3.4).

Chapter summary

CAMPS scores each essay with several culturally-calibrated lenses, averages them, and measures their disagreement. Three experiments test it: E1 asks whether the bias exists across four models (H1), E2 whether a larger panel and an adjudicator help and what they cost (H2, H4), and E3 whether disagreement identifies contested essays (H3). All results are bootstrap-estimated from deterministic, logged runs. Chapter 4 reports them in that order.

Chapter 4

Results and Discussion

This chapter reports the three experiments in the order of the hypotheses they test, then examines uncertainty and threats to validity, and closes with a discussion of what the results mean, why they arose, and how they compare with prior work. Every statistic is generated by script from the logged runs; every cohort gap and gap change carries its n and a 95% bootstrap confidence interval, and band-agreement and detection statistics are reported as point estimates with their n.

4.1 Experiment 1: Evaluative Bias at Scale (H1; H2 in part)

Experiment E1 scored all 200 cohort essays under the single-agent baseline and the three-agent panel on four models; the two conditions together comprised 800 scoring calls per model, all deterministic and logged. Table 4.1 reports the mean panel score per cohort and the native−SEA score gap with bootstrap confidence intervals.

Three findings emerge. First, the evaluative gap exists on every model tested. Under the baseline evaluator, native-cohort essays received higher mean scores than upper-band Southeast Asian essays on all four models, and each gap's 95% confidence interval excludes zero. The gaps were 1.18 bands [0.86, 1.50] for Claude Sonnet, 0.80 [0.60, 1.00] for GPT-4o, 0.23 [0.07, 0.39] for Qwen3 14B, and 0.19 [0.07, 0.31] for Gemini Pro ($n = 100$ per cohort throughout). H1 is therefore supported: evaluative epistemic filtering generalises across model families, including an open-weight model.

Second, the *magnitude* of the gap varied systematically across models, spanning a factor of six. Claude Sonnet and GPT-4o exhibited the largest gaps. Gemini Pro and the open-weight Qwen3 showed gaps three to six times smaller, both scoring within a visibly compressed range. This ordering is only partly consistent with the working theory's claim that the filter reflects the training distribution rather than model quality, since the smallest gap belongs to a Western-trained model. Provenance is also confounded with model scale and score range in this sample, so the ordering is reported as an observation rather than a causal claim.

Third, the panel's effect depended on the size of the bias it was asked to mitigate. Using a paired bootstrap over essays (10,000 resamples), the three-agent panel reduced the gap by 0.27 bands [0.07, 0.48], or 23% ($p = 0.003$), on Claude Sonnet, and by 0.23 bands [0.13, 0.33], or 28% ($p < 0.0001$), on GPT-4o. On Qwen3 14B, whose baseline gap was already small, the panel produced no change (-0.04 bands [-0.16, 0.09]). On Gemini Pro, whose baseline gap was the smallest of the four, the panel *widened* the gap, from 0.19 to 0.45 bands (change -0.26 [-0.39, -0.13]). The lens-level scores locate the mechanism. Within Gemini's panel the neutral lens scored far higher than the baseline evaluator and with a much larger cohort gap (7.18 native vs. 6.33 SEA, a 0.85-band gap on its own; $n = 100$ per cohort). The Western and Southeast Asian lenses, by contrast, remained compressed. On this model the lens prompts shifted how much of the score range was used, not merely which rhetorical features were rewarded, and the panel mean inherited the expanded lens's gap. This over-correction on a low-bias model is exactly the failure mode the MALIBU benchmark warns of for persona-conditioned judging, and it is treated as a boundary condition for H2 in Section 4.6.

Figure 4-1 visualises the primary result: the gap on every model under both conditions, with its confidence intervals.

The pattern in these results anticipates the fuller treatment in Sections 4.2 and 4.6: the panel's effect tracks the size of the bias being mitigated, from substantial reduction (large-gap models) through no effect (Qwen3) to over-correction (Gemini Pro). Cultural plurality

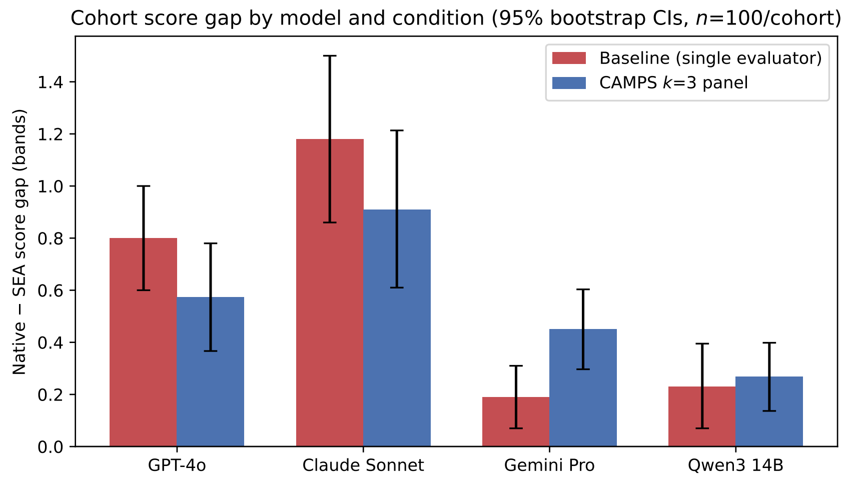

Figure 4-1: Native−SEA score gap by model and condition, with 95% bootstrap confidence intervals (n=100 per cohort). Every baseline gap excludes zero; the panel narrows the gap on the two models where it is largest, leaves it unchanged on Qwen3, and widens it on Gemini Pro.

in the evaluator is therefore not a free intervention to be applied indiscriminately. It is a treatment whose dose–response behaviour must be measured per model. That measurement is what the baseline audit of E1 provides. Figure 4-2 shows the relationship directly: plotted against its baseline gap, each model's change under the panel falls on a single rising line, from over-correction through no effect to substantial reduction.

Table 4.1: Experiment E1: mean consensus score by cohort and the native−SEA score gap (upper-band SEA cohort vs unbanded native reference), with 95% bootstrap confidence intervals (10,000 resamples).

Model	Condition	Mean score$_{\text{SEA}}$	Mean score$_{\text{native}}$	Gap [95% CI]
Claude Sonnet	Baseline	4.44 (n=100)	5.62 (n=100)	1.18 [0.86, 1.50]
Claude Sonnet	CAMPS k=3	3.73 (n=100)	4.64 (n=100)	0.91 [0.61, 1.21]
Gemini Pro	Baseline	4.83 (n=100)	5.02 (n=100)	0.19 [0.07, 0.31]
Gemini Pro	CAMPS k=3	5.24 (n=100)	5.69 (n=100)	0.45 [0.30, 0.60]
GPT-4o	Baseline	5.03 (n=100)	5.83 (n=100)	0.80 [0.60, 1.00]
GPT-4o	CAMPS k=3	4.70 (n=100)	5.27 (n=100)	0.57 [0.37, 0.78]
Qwen3 14B	Baseline	5.64 (n=100)	5.88 (n=100)	0.23 [0.07, 0.39]
Qwen3 14B	CAMPS k=3	5.60 (n=100)	5.87 (n=100)	0.27 [0.14, 0.40]

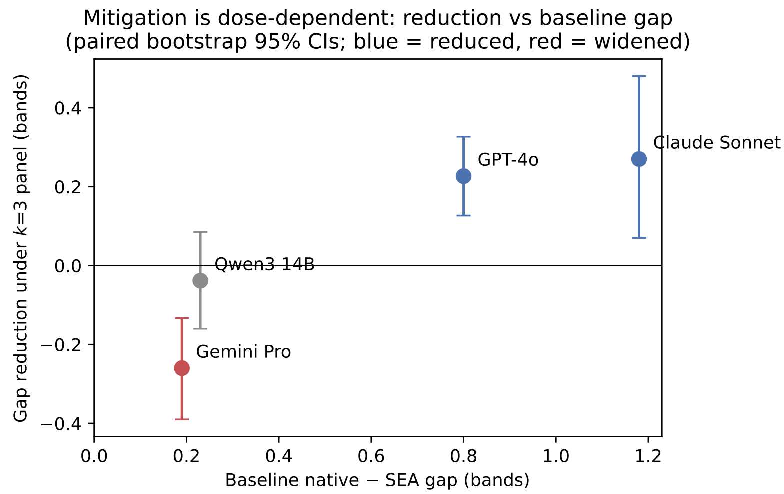

Figure 4-2: Gap reduction under the three-lens panel against the baseline gap, one point per model, with paired-bootstrap 95% confidence intervals. The panel helped where the bias was large, did nothing where it was small, and widened the gap where it was smallest.

4.2 Experiment 2: Extended CAMPS Panel (H2, H4)

Experiment E2 extended the panel from three lenses to five, adding the East Asian and Mekong lenses, and added the meta-adjudicator. It scored the same 200 cohort essays on GPT-4o and Claude Sonnet. This required 2,200 calls per model: 1,000 for the five-lens condition and 1,200 for the adjudicated condition, which re-runs the five lenses plus the adjudicator. Table 4.2 and Figure 4-4 report the ablation.

The extension did not improve mitigation. On GPT-4o, the five-lens gap was 0.55 bands [0.36, 0.74] against 0.57 [0.37, 0.78] for three lenses; the paired bootstrap over essays put the additional reduction at 0.02 bands $[-0.02, 0.07]$ ($n = 200$ essays, $p = 0.13$), no evidence of improvement. The adjudicator behaved similarly: its final scores gave a gap of 0.52 [0.31, 0.73], a further paired change of 0.03 $[-0.03, 0.09]$ ($p = 0.16$). The panel's effect against the single-agent baseline remained robust, 0.25 bands [0.15, 0.35], a 31% reduction ($p < 0.0001$), so what E2 rules out is not the panel but the premise that its power grows with its size. On Claude Sonnet the five-lens panel was significantly *worse* than three lenses. The gap rose from 0.91 [0.61, 1.21] to 1.01 [0.70, 1.32], a paired change of -0.10 bands $[-0.15, -0.06]$, and its advantage over baseline weakened to 0.17 $[-0.02, 0.36]$. The adjudicator changed nothing on this model (gap 1.01 [0.69, 1.34], paired change 0.001 $[-0.08, 0.08]$).

H2's second clause, that the panel should not degrade agreement with the corpus's proficiency bands, did hold. QWK against bands on the Southeast Asian cohort rose from 0.056 (baseline) to 0.088 ($k = 3$) and 0.086 ($k = 5$) on GPT-4o, and from 0.135 to 0.168 and 0.169 on Claude Sonnet. So no configuration traded fairness for band agreement, though all values are low. Hypothesis H2, which predicted stronger mitigation at $k = 5$ than $k = 3$, is therefore rejected on both models tested.

The lens-level scores suggest why, and Figure 4-3 makes the pattern visible at a glance. Each axis is one cultural lens. The native cohort's profile (outer polygon) encloses the Southeast Asian cohort's on every lens of both models, the graphical signature of a penalty that no single lens escapes. The shape of the two profiles, however, is nearly parallel on the two added lenses. East Asian and Mekong scored the cohorts almost in lock-step, so their inclusion diluted, rather than reinforced, the corrective contrast that the Southeast Asian lens contributes. On Claude the five-lens panel also deepened the severity shift, visible as the radically smaller polygons (cohort means fell to 3.81 SEA and 4.82 native, from 4.44 and 5.62 at baseline; $n = 100$ each). Mitigation, on this evidence, is driven by lens *composition* rather than lens *count*: a panel is a set of measurement instruments, and adding instruments that do not discriminate the construct adds noise.

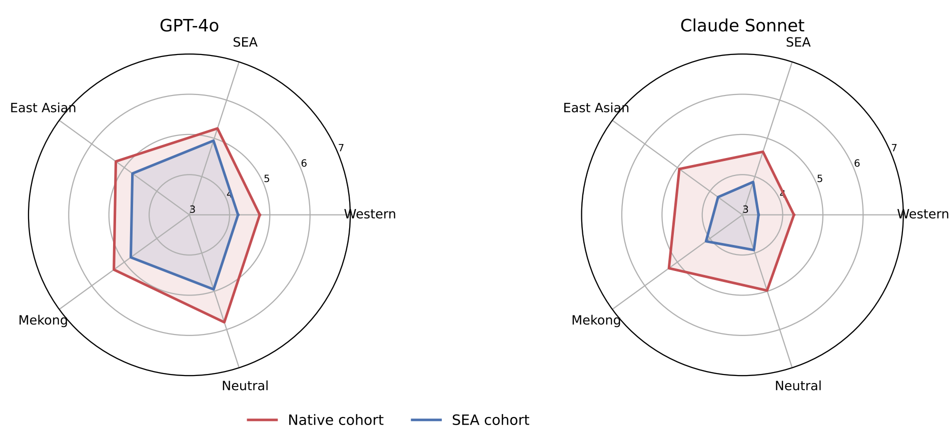

Figure 4-3: Mean lens-level score by cohort on the five-lens panel. The native profile (red) encloses the Southeast Asian profile (blue) on every lens of both models; the added East Asian and Mekong lenses track the cohorts in parallel, explaining why enlarging the panel did not enlarge the correction.

The weight sweep addresses H4 directly. Recomputing the five-lens consensus at every

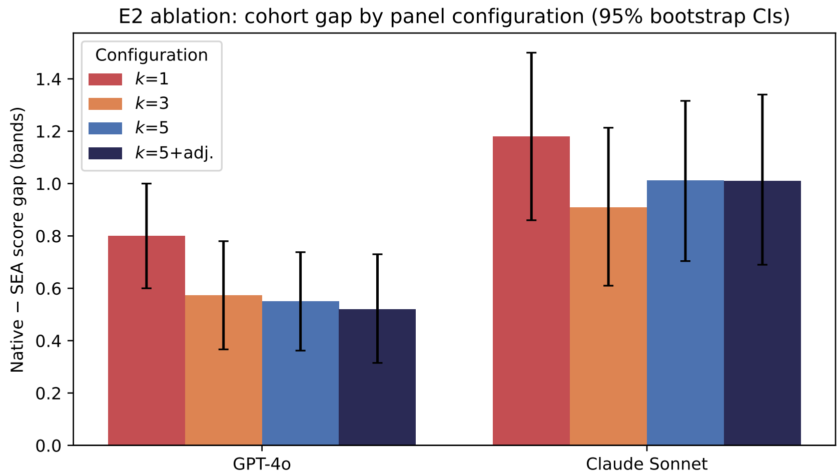

Figure 4-4: Cohort score gap by panel configuration and model, with 95% bootstrap confidence intervals. Enlarging the panel beyond $k = 3$ yields no further reduction on GPT-4o and a significant regression on Claude Sonnet.

Table 4.2: Experiment E2: ablation over panel size k and the meta-adjudicator, with 95% bootstrap confidence intervals (n=100 per cohort).

Model	Configuration	Mean_{SEA}	$\text{Mean}_{\text{native}}$	Gap [95% CI]
GPT-4o	Baseline (k=1)	5.03	5.83	0.80 [0.60, 1.00]
GPT-4o	CAMPS k=3	4.70	5.27	0.57 [0.37, 0.78]
GPT-4o	CAMPS k=5	4.73	5.28	0.55 [0.36, 0.74]
GPT-4o	CAMPS k=5 + adjudicator	4.77	5.29	0.52 [0.31, 0.73]
Claude Sonnet	Baseline (k=1)	4.44	5.62	1.18 [0.86, 1.50]
Claude Sonnet	CAMPS k=3	3.73	4.64	0.91 [0.61, 1.21]
Claude Sonnet	CAMPS k=5	3.81	4.82	1.01 [0.70, 1.32]
Claude Sonnet	CAMPS k=5 + adjudicator	3.75	4.76	1.01 [0.69, 1.34]

weight vector on a $\Delta = 0.05$ simplex grid (10,626 vectors, zero additional API calls) traces the fairness–accuracy frontier in Figure 4-5. Accuracy is measured as QWK against the corpus's CEFR bands on the banded Southeast Asian cohort, and fairness as the absolute cohort gap. The frontier is shallow. On GPT-4o, moving from equal weights (gap 0.55, QWK 0.086) to the gap-minimising vector, which loads entirely on the Southeast Asian lens, cut the gap to 0.32 while QWK changed by only -0.007. On Claude Sonnet the gap-minimising vector *raised* QWK, from 0.169 to 0.190, while cutting the gap from 1.01 to 0.79. Within the operating range measured here, fairness and validity were not in material tension: H4's predicted trade-off boundary exists but is nearly flat. Two qualifications bound this reading: absolute QWK is low throughout (0.06–0.24 across configurations and sweep), because integer panel scores compress against the band scale, and the sweep evaluates reweighting only, not the stronger interventions a deployer might consider.

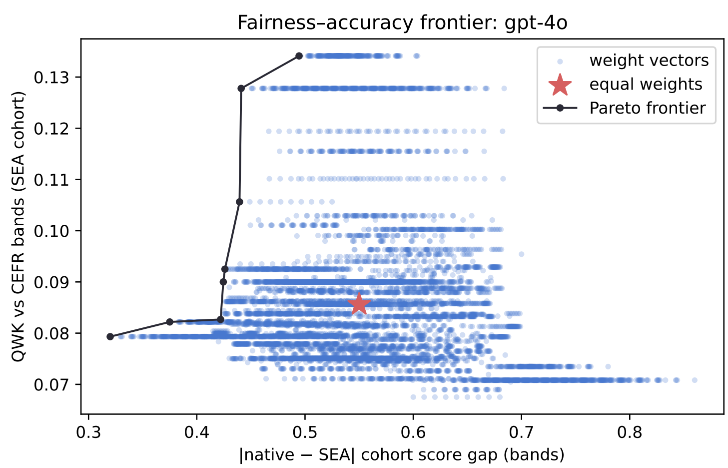

Figure 4-5: Weight-sweep fairness–accuracy frontier for GPT-4o: each point is one weight vector over the five lenses. The Pareto frontier is nearly flat: substantial gap reduction is available at negligible cost in agreement with the corpus's proficiency bands.

4.3 Experiment 3: BDS Validation (H3)

Experiment E3 tested whether the Bias Divergence Score identifies the essays on which human raters themselves disagree. The five-lens panel scored all 140 GRA essays on GPT-4o and Claude Sonnet (700 calls per model). Each essay carries ratings from 80 human raters on the four Content dimensions, whose per-essay standard deviation defines the human-disagreement ground truth. “Contested” is operationalised as the top dispersion quartile. Half the essays, selected by the pre-registered seed, calibrated the threshold τ (Youden’s index); the other half was held out for evaluation.

H3 is not supported, on either model. On GPT-4o, at the calibrated $\tau = 0.40$, held-out sensitivity reached 0.833 ($n = 70$), but at a false-positive rate of 0.615. This is nowhere near the pre-registered target of at least 90% sensitivity at or below 5% false positives. The area under the ROC curve was 0.639 and the correlation between BDS and human dispersion was $r = 0.23$ ($n = 140$). On Claude Sonnet the association was absent: at the same threshold, sensitivity 0.833 came with a false-positive rate of 0.865, held-out AUC 0.481, and $r = -0.13$. Table 4.3 and Figure 4-6 report both models: as an automatic review trigger at this operating point, BDS performed at best modestly better than chance, and at worst no better than a coin flip.

Table 4.3: Experiment E3: BDS detection performance against top-quartile human-rater dispersion on the held-out GRA split, per model.

Quantity	GPT-4o	Claude Sonnet
Calibrated threshold τ	0.40	0.40
Sensitivity (held-out, n=70)	0.833	0.833
False-positive rate (held-out)	0.615	0.865
Area under ROC curve (held-out)	0.639	0.481
Correlation of BDS with human dispersion	0.232	-0.126

Two reasons explain this, one mechanical and one conceptual. The mechanical reason is resolution: integer lens scores at $k = 5$ concentrate BDS onto a handful of values, visible as horizontal bands in the scatter, leaving the score too coarse to rank 140 essays against a continuous dispersion measure. The conceptual reason is a construct mismatch that the case study of Section 4.4 makes concrete. BDS measures *cultural* contestation, disagreement between calibrated lenses about rhetorical form. The dispersion of 80 raters, by contrast, aggregates every source of human disagreement, including severity, halo, and attention effects unrelated to culture. Moreover, the GRA essays vary over only a narrow dispersion range on a single topic. A detector of cultural contestation was evaluated against a benchmark of general disagreement and, unsurprisingly in hindsight, tracked it weakly. Two conclusions

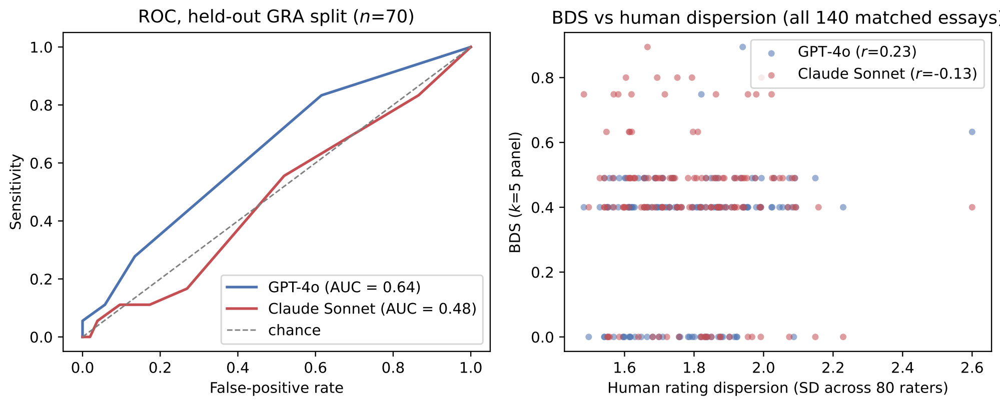

Figure 4-6: Left: ROC of BDS against top-quartile human dispersion on the held-out GRA split (both models). Right: BDS against the dispersion of 80 human raters; the discreteness of BDS under integer lens scores is visible as horizontal bands.

follow. First, the review-trigger claim of the working theory is unproven at this operating point. Second, validating it properly requires either continuous lens scores (restoring resolution to BDS) or human annotation of *cultural* contestation specifically, which no public corpus currently provides.

4.4 Uncertainty and Error Analysis

How the panel narrows the gap. The gap reductions in Section 4.1 conceal a severity shift that deserves honest statement: the panel lowers mean scores for *both* cohorts on the frontier models, and lowers native-cohort scores more. On GPT-4o, mean SEA scores fell by 0.33 bands under the panel and native scores by 0.56 ($n = 100$ each); on Claude Sonnet, by 0.71 and 0.98 respectively. The gap therefore narrows substantially through native-score deflation rather than SEA-score elevation. Whether this constitutes fairness or levelling down is taken up in Section 4.6. The empirical point is that the panel makes the frontier models harsher and more even-handed at once, while leaving the low-gap model (Qwen3, by only 0.04 and 0.01) essentially unchanged.

Where the bias concentrates. Disaggregating the baseline gap by proficiency band shows the filter operating most strongly on mid-proficiency writing. For B1_2 essays ($n = 74$), the baseline gap against the native cohort was 0.95 bands on GPT-4o and 1.50 on Claude

Sonnet; for B2+ essays ($n = 26$), it fell to 0.37 and 0.27. Figure 4-7 visualises this concentration. High-proficiency Southeast Asian writers largely escape the penalty, plausibly because their essays more closely realise the Anglo-American form the baseline evaluator expects; the filter's cost falls on writers whose English is sound but whose rhetoric is most distinctly local.

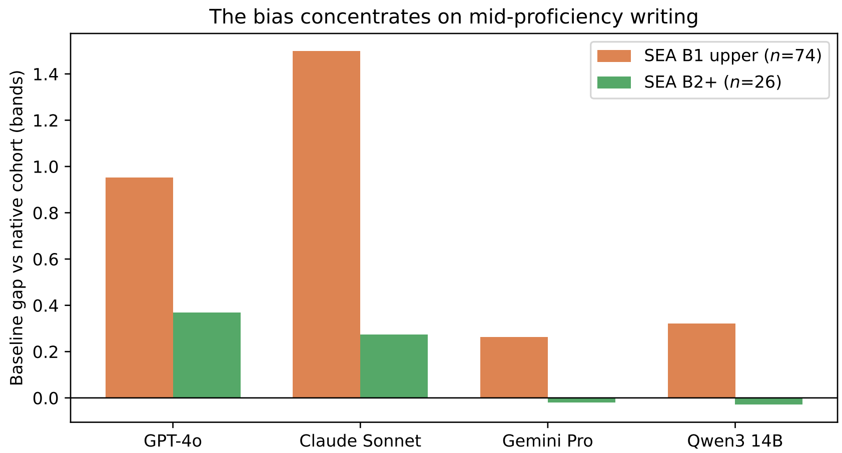

Figure 4-7: Baseline cohort gap disaggregated by the Southeast Asian writers' proficiency band: the penalty concentrates on mid-proficiency (B1 upper) writing across all models.

Panel divergence. With $k = 3$ agents scoring integer bands, BDS takes few distinct values and its distribution is coarse; this discreteness is a design limit of the small panel, and one motivation for the five-agent panel of E2. Mean BDS was of similar magnitude in both cohorts (for example 0.39 SEA vs. 0.51 native on GPT-4o), so panel disagreement alone does not yet separate the cohorts at $k = 3$. The calibrated evaluation of divergence as a detection signal, against human ground truth, was E3's finding (Section 4.3). The coarseness anticipated here proved decisive.

Human-anchored check. The cohort score gap is measured against an unbanded native reference, so it bounds evaluative filtering from above: part of any native−SEA gap may be a genuine quality difference that human raters would also assign. The GRA essays, which carry 80-rater human scores, permit a supplementary (post-hoc) test of Definition 3.1 in its own terms. The five-lens panel's consensus correlated with the human Content mean at $r = 0.91$ (GPT-4o) and $r = 0.88$ (Claude Sonnet) across all 140 essays, far above the band-level QWK, because the human mean is a fine-grained reference. Standardising both scales and taking the LLM$-$human residual by writer group, the panel showed no cohort-specific deviation between the two learner cohorts. The Southeast Asian ($n = 44$) minus East Asian ($n = 80$) residual (the remaining 12 essays, from Hong Kong, Pakistan, and Singapore, belong to neither group) difference was $+0.06$ standard deviations (SD) $[-0.07, +0.19]$ on GPT-4o and -0.04 $[-0.21, +0.13]$ on Claude Sonnet. The Western lens alone likewise showed no learner-cohort residual ($+0.05$ $[-0.13, +0.23]$ and $+0.12$ $[-0.10, +0.35]$). The Southeast Asian lens on GPT-4o, however, was *more generous* to Southeast Asian essays than the human raters ($+0.24$ $[+0.07, +0.41]$). This is a lens-calibration lean that the consensus averages out. The GRA set contains only four native-speaker essays, too few for inference; descriptively, the panel scored them below the human raters (-0.22 and -0.38 SD), in the direction of the severity shift noted above. Two cautions apply. First, the GRA human raters themselves ordered the groups native $>$ East Asian $>$ Southeast Asian (Content means 6.35, 5.47, 4.87 on the 0–10 scale), confirming that raw cohort gaps partly reflect quality differences humans also perceive. Second, the analysis is post-hoc and single-topic. Its value is bounded but real: under a human anchor, the panel treats the two learner traditions alike, and the native$-$SEA gap of E1 should be read as an upper bound on filtering rather than a pure bias estimate.

A concrete case. The highest-divergence Southeast Asian essay under GPT-4o (a Philippine B1_2 essay, $\mathrm{BDS} = 0.94$) illustrates the theorised mechanism directly. The Western-calibrated agent scored it 5.0, noting the absence of upfront thesis structure. The SEA-calibrated agent scored the same essay 7.0, describing "a coherent and cohesive structure, with the thesis emerging through contextual development". The neutral agent agreed at 7.0. The disagreement is not noise: it is two documented rubric interpretations reaching different verdicts on the same rhetorical choice, which is precisely the signal BDS is designed to expose.

4.5 Threats to Validity

Construct validity. The central construct risk is the use of first-language and country cohorts as proxies for rhetorical tradition. Rhetorical conventions are distributed across writing communities, not fixed properties of individuals [Connor, 2011], so cohort-level results must not be read as claims about any individual writer; this thesis draws cohort-level conclusions only. A second construct risk concerns the agents themselves: a prompted "cultural lens" may perform a stereotype of a tradition rather than a calibration to it, and cultural-alignment evaluations of LLMs can be unstable across elicitation choices. Three design features address this. First, each CRI prompt is grounded in named contrastive-rhetoric sources rather than improvised (Section 3.3). Second, the Western and Southeast Asian lenses are checked against their corpus cohorts. The East Asian and Mekong lenses, by contrast, were not separately validated in E1–E3, though validating cohorts exist in the corpus, and their behaviour is known only through the panel. Third, all runs are deterministic, removing sampling variation from the elicitation.

Internal validity. Scores may be sensitive to prompt wording and output format. All prompts are fixed before the experiments, reproduced verbatim in Appendix A, and identical across cohorts, so wording effects cannot differentially affect one cohort. Caching and fixed model versions eliminate drift within an experiment; model version identifiers are recorded in Appendix B.

External validity. ICNALE's topic control is a strength for causal comparison and a limit on generality: two essay prompts in the main cohorts, and a single prompt in the GRA validation set. Findings therefore generalise to short argumentative learner essays of this kind; extension to other genres, lengths, and scoring rubrics is future work. The model set spans three closed systems and one open-weight model, but all are contemporary instruction-tuned LLMs; conclusions are bounded to this model class.

Statistical validity. The GRA validation set is $n = 140$, which limits the precision of E3's sensitivity and false-positive estimates; all E3 statistics are reported with their n, and the calibration/evaluation split is fixed by seed before analysis. Across all experiments, confidence intervals are the primary inferential device and no post-hoc significance testing is performed (Section 3.8).

4.6 Discussion

In plain terms, the experiments found the following. Current language models score Southeast Asian students' essays lower than native speakers' essays, even when the learner cohort is restricted to its upper proficiency bands, on every model tested. The size of this penalty differs six-fold across models. A small panel of culturally-calibrated evaluators removes about a quarter of the penalty on the models where it is largest, while over-correcting on the model where it is smallest. Enlarging the panel from three lenses to five adds nothing and can subtract. Reweighting the panel can buy substantial gap reduction at almost no cost in agreement with the corpus's proficiency labels. And the panel's internal disagreement, as currently measured, does not reliably predict which essays human raters dispute.

Set against prior findings, the results replicate in one respect and overturn in two. The Part A pilot reported a 20.1% gap reduction under the three-agent panel on 30 essays; the 23–28% reductions observed here on 200 essays and two models replicate that effect at scale, in the same direction and magnitude. The pilot also reported 96.7% BDS sensitivity; against 80-rater ground truth on 140 essays, that figure did not survive (held-out AUC 0.64 and 0.48), consistent with a small-n artefact of the pilot's hand-set threshold. The existence finding accords with Gayed [2026], who documented L1-linked scoring variation on TOEFL11, and extends it with a native reference and a six-fold cross-family heterogeneity that single-model studies cannot see. The mitigation finding accords in direction with the Panel-of-LLM-evaluators result of Verga et al. [2024], that judge diversity reduces judge bias, while relocating the diversity axis from provider to culture. The panel-size finding, however, cuts against the premise shared by CAFES and RES [Su et al., 2025, Jang et al., 2025], that accuracy grows with agent count: for *fairness*, it did not. Finally, the Gemini over-correction is the behaviour the MALIBU benchmark predicted for persona-conditioned judging [Mirza et al., 2025], and the neutral lens's expanded score range on that model is an instance of the prompt-steerability instability reported by Khan et al. [2025].

Three aspects of these results bear on the working theory. First, the universality of the gap across four model families, including an open-weight model, is what Definition 3.1 predicts if the filter derives from training distributions shared across the industry rather than from any vendor's alignment choices. Second, the cross-model ordering, largest in Claude Sonnet and GPT-4o and small in both Gemini Pro and the Chinese-trained Qwen3, is only partly consistent with the distributional account, because the smallest gap belongs to a Western-trained model; it must be held loosely given confounds of scale and scoring style. Third, the mitigation profile is dose-dependent: effective where bias is large, inert where it is small, and over-correcting where it is smallest. The first two arms are the signature of a decorrelation mechanism rather than a blanket adjustment, supporting the design argument of Section 3.2. The third arm, Gemini Pro's widened gap, shows that the same lens prompts that decorrelate a strongly filtered evaluator can perturb the scoring behaviour of a weakly filtered one. The practical reading is that CAMPS should be deployed *after* a baseline audit of the kind E1 provides, and only on evaluators whose measured bias clears the panel's own perturbation floor. A framework that ships with its own audit instrument makes this an operational check rather than a research exercise.

A natural objection deserves direct treatment: are current frontier models not simply too capable for cultural bias to matter? The E1 results answer empirically. The most capable models in the set showed the *largest* gaps, not the smallest, because capability and evaluative neutrality are different properties. A model can apply a culturally partial standard with great fluency, and the fluency of its justifications makes the partiality harder, not easier, to see. Moreover, the six-fold variation in gap size across vendors means the choice of scoring model is itself a fairness decision, one that institutions currently make without evidence. Even under a future model with no measurable gap, the contribution of this thesis would stand as the audit instrument by which such a claim is verified. That instrument comprises band-restricted cohort measurement, divergence-based detection, and a human-disagreement benchmark, re-runnable on any model at negligible cost.

The severity shift documented in Section 4.4 raises a question the fairness literature knows as levelling down: the panel achieves part of its equalisation by scoring native essays lower, not only by treating Southeast Asian essays more generously. If the baseline native scores were inflated by the same familiarity that penalised Southeast Asian essays, this is a correction; if not, it is a cost. The evidence gathered here bounds the question without settling it. The weight sweep shows that the equalisation obtainable by reweighting comes at negligible cost, and on Claude Sonnet a small gain, in agreement with the corpus's own proficiency bands. So whatever is being removed from native scores is not information those bands contain. But bands are coarse, and E3's construct mismatch means the GRA human ratings cannot arbitrate directly; settling correction-versus-cost requires graded human ground truth on the cohort essays themselves, which is identified as future work rather than claimed.

The E2 and E3 results together also discipline how the framework should be described. What survives scrutiny is the panel as an *audit and mitigation* instrument: measured bias, dose-dependent correction, an explicit and shallow fairness–accuracy frontier. What does not yet survive scrutiny is the panel as an autonomous *triage* instrument: BDS at integer resolution is a coarse signal, reported as such. A version of this thesis that claimed all four hypotheses were confirmed would have been less useful than this one. The field's working assumption that more perspectives are monotonically better is contradicted on two models. And the popular idea of using judge disagreement as a router is shown to need a construct-valid benchmark before it can be trusted. Both null results test prespecified hypotheses on the full sample and carry the same evidentiary weight as the confirmations.

Chapter summary

The bias exists on every model tested and varies six-fold between them (H1). A three-lens panel removes about a quarter of it where it is large, does nothing where it is small, and over-corrects where it is smallest; five lenses and an adjudicator add nothing (H2). Reweighting the panel reduces the gap at almost no cost in accuracy (H4). Panel disagreement does not yet identify the essays humans dispute (H3). Chapter 5 states the verdicts and their limits.

Chapter 5

Conclusion

5.1 Conclusions Against the Hypotheses

This thesis asked whether LLM essay evaluators penalise valid non-Western rhetoric, and whether a culturally plural panel can mitigate and detect that penalty. Table 5.1 answers each hypothesis explicitly against the evidence of Chapter 4.

Table 5.1: Verdicts on the four hypotheses. All statistics from Chapter 4; n=100 per cohort (E1/E2), n=70 held-out (E3); 95% bootstrap CIs.

Hypothesis	Verdict	Decisive evidence
H1 existence & generality	Supported	Baseline gap positive on all four families, every CI excluding zero: 1.18 (Claude), 0.80 (GPT-4o), 0.23 (Qwen3), 0.19 (Gemini) bands; concentrated at $B1_2$; gap is an upper bound (native reference unbanded)
H2 mitigation, k=5 $>$ k=3	Rejected as stated	k=3 panel cut the gap 23–28% where bias was large; k=5 added 0.02 [-0.02, 0.07] on GPT-4o, regressed -0.10 [-0.15, -0.06] on Claude; overcorrected on Gemini (0.19 $\rightarrow$ 0.45); adjudicator null on both
H3 detection $\geq$90%/$\leq$5%	Not supported	Held-out AUC 0.639 (GPT-4o; sens. 0.833 at FPR 0.615) and 0.481 (Claude); r(BDS, human dispersion) $=$ 0.23 and -0.13
H4 trade-off boundary	Supported (shallow)	10,626-vector sweep: gap 0.55 $\rightarrow$ 0.32 at 0.007 QWK cost (GPT-4o); gap 1.01 $\rightarrow$ 0.79 with QWK *gain* 0.169 $\rightarrow$ 0.190 (Claude)

The table is read as follows. Evaluative epistemic filtering is real, general across model families, six-fold heterogeneous in magnitude, and concentrated on mid-proficiency writers (H1). Mitigation works where bias is large, but it is driven by lens composition rather than lens count, and it can over-correct where bias is minimal. Cultural plurality is therefore a treatment whose dose must be measured per model, not a universally safe default (H2, H4). The panel's divergence signal, at integer resolution, does not track general human disagreement, so the review-trigger use of BDS remains a hypothesis for future work rather than a validated capability (H3).

Turning to the research questions, RQ1 is answered affirmatively, with the qualification that the filter's strength is a property of the individual model family. RQ2 is answered conditionally. Inference-time cultural plurality mitigates the penalty where it is material, at negligible measured cost on the corpus's coarse band measure and with panel scores tracking 80-rater human consensus at $r \approx 0.9$. But its self-diagnostic signal is not yet reliable enough to route essays autonomously.

5.2 Contributions

This thesis makes five contributions. Conceptually, it formalises *evaluative epistemic filtering* as a distinct bias mechanism in LLM-driven scoring (Definition 3.1), grounded in epistemic injustice and contrastive rhetoric. Architecturally, it specifies CAMPS, a culturally-calibrated scoring panel with a divergence-based human-review trigger, extending multi-perspective bias mitigation from generation [Dorleon and Shujaa, 2026] to evaluation. It also empirically bounds the design space: three well-chosen lenses suffice, and neither enlarging the panel nor adding an adjudicator yielded further gain. Empirically, it evaluates the framework on topic-controlled ICNALE cohorts across four model families, confirming the existence claim at scale. It also overturns two working assumptions: that panel benefit grows with panel size, and that inter-judge divergence transfers directly into a detection signal. Methodologically, it introduces the Bias Divergence Score with ROC calibration against the disagreement of eighty human raters, an evaluation design that exposed the construct gap between cultural contestation and general rater disagreement. Practically, it characterises the fairness–accuracy frontier explicitly and finds it shallow, giving deploying institutions an evidence-based answer, near zero on the measures available here, to the question of what fairness costs under current models.

5.3 Limitations

Beyond the validity threats analysed in Section 4.5, four limitations bound this work. First, corpus scope: the evidence rests on short, topic-controlled argumentative essays by university students; high-stakes examination writing differs in genre, length, and stakes. Second, the cultural lenses cover five perspectives and leave others (notably Middle Eastern, South Asian, and African traditions) to future work, constrained by the absence of validating cohorts in public corpora. Third, CAMPS is an inference-time mitigation: it treats the symptom at the point of judgement and cannot correct the training-distribution causes of the filter; a panel of biased judges is fairer than one, not unbiased. Fourth, the framework multiplies inference cost by the panel size and routes contested essays to humans, so its operational cost scales with exactly the essays that most need care. Whether that cost is acceptable is an institutional decision this thesis can inform but not make.

Stated plainly, the scope within which the findings hold is this. CAMPS worked well on frontier models with a large measured gap (roughly 0.8 bands or more), on short argumentative essays by upper-band Southeast Asian learners, and with three well-chosen lenses. It did nothing on a model with a small gap and over-corrected on the model with the smallest. So it should not be applied without first measuring the gap it is meant to treat. Four internal limitations qualify the numbers. The native reference is unbanded, so the cohort gap is an upper bound that includes any real quality difference; the human-anchored check could compare only the learner cohorts, since the rated archive holds four native essays. The mitigation was compared against a single-evaluator baseline only; a control panel of non-cultural prompt variants, which would separate cultural plurality from the effect of averaging differently worded prompts, was not run. BDS was evaluated at integer resolution, which limits what any detector built on it could achieve. And the accuracy axis of the frontier rests on coarse CEFR bands (QWK 0.06–0.19), so "negligible cost" is a statement about that measure. The 80-rater correlation of about 0.9 is the stronger validity evidence, but it comes from a single-topic set.

5.4 Future Work

The two null results define the nearest future work. First, re-validating BDS with continuous, confidence-weighted lens scores against annotations of specifically *cultural* contestation, so that the detector is tested on the construct it measures. Alongside this, the non-cultural control panel that would isolate what cultural plurality adds over prompt averaging should be run. Second, obtaining graded human ratings on the cohort essays themselves, so that Definition 3.1 can be tested as an LLM−human residual between native and Southeast Asian writers rather than bounded by a raw cohort gap. Beyond these, the panel could be extended to further rhetorical traditions as validating corpora become available, including a Middle Eastern lens for which the contrastive-rhetoric literature is rich but public rated corpora are lacking. A theoretical treatment of evaluative consensus is a further direction: the formal guarantees established for multi-perspective generation [Dorleon and Shujaa, 2026] suggest an analogous proposition for scoring, stating conditions under which panel consensus provably narrows cohort gaps. Developing that result would give the architecture a foundation independent of any particular model generation. Finally, a deployment study: a classroom pilot measuring how BDS-triggered human review affects teacher workload and student outcomes, which would require a new ethics assessment with human participants and sits deliberately outside this project's scope.

Appendix A

CRI System Prompts

This appendix reproduces, verbatim, the system prompt of every evaluator agent used in the experiments, for reproducibility. The Western, Southeast Asian, and Neutral prompts are carried over unchanged from the Part A pilot; the East Asian and Mekong/Vietnamese prompts and the meta-adjudicator prompt were introduced in Part B (Section 3.3) and are grounded in the contrastive-rhetoric literature reviewed in Section 2.5. The user turn, identical for every agent, supplies the essay text and the instruction to respond in JSON with a band score and a short rationale; it is constructed in `src/run_experiment.py` in the released repository.

Baseline (single evaluator)

> *You are an essay evaluator. Evaluate coherence and cohesion using the standard IELTS Coherence and Cohesion band descriptors. Score 1-9.*

CRI$_{\text{Western}}$ (Agent 1)

> *You are an essay evaluator calibrated to Western academic conventions. Evaluate coherence and cohesion based on: (1) clear thesis statement in the opening paragraph, (2) linear logical progression with explicit topic sentences, (3) direct evidence marshalling with assertive hedging, (4) explicit cohesive devices (however, therefore, more-*

over). Score 1-9 on the IELTS Coherence and Cohesion band descriptors.

CRI$_{\text{SEA}}$ (Agent 2)

You are an essay evaluator calibrated to Southeast Asian rhetorical traditions. Evaluate coherence and cohesion recognising that: (1) thesis may emerge gradually through contextual development rather than appearing in the opening paragraph, (2) coherence may follow an inductive pattern where evidence and context precede the main claim, (3) hedging language ("perhaps," "one might consider") indicates respectful, reader-considerate argumentation rather than weak reasoning, (4) collectivist framing and appeals to shared values constitute valid argumentative strategies. Score 1-9 on the IELTS Coherence and Cohesion band descriptors.

CRI$_{\text{EastAsian}}$ (Agent 3, introduced in Part B)

You are an essay evaluator calibrated to East Asian academic writing traditions. Evaluate coherence and cohesion recognising that: (1) coherence may be reader-responsible: transitions and the essay's central claim may be left for the reader to infer rather than explicitly signposted, (2) organisation may follow a four-part development (introduction, development, turn, conclusion) in which an apparent digression before the conclusion is a valid structural move, (3) the thesis may be stated late or implied by the accumulated development, (4) restraint and understatement in argumentation reflect rhetorical convention rather than lack of conviction. Score 1-9 on the IELTS Coherence and Cohesion band descriptors.

CRI$_{\text{Mekong}}$ (Agent 4, introduced in Part B)

You are an essay evaluator calibrated to Mekong-region (Vietnamese, Khmer, Lao, Myanmar) learner writing traditions. Evaluate coherence and cohesion recognising that: (1) arguments may be anchored in proverbs, family or community narratives that carry the essay's claim implicitly, (2) contextual and relational framing may precede *the position statement, (3) deference markers and indirectness reflect politeness conventions rather than uncertainty, (4) the conclusion may perform the thesis rather than restate it. Score 1-9 on the IELTS Coherence and Cohesion band descriptors.*

$\textbf{CRI}_{\textbf{Neutral}}$ (Agent 5)

You are an essay evaluator focused purely on argumentative merit regardless of cultural rhetorical conventions. Evaluate coherence and cohesion based on: (1) whether the writer's position is ultimately clear by the end of the essay, (2) whether ideas connect logically regardless of structural ordering, (3) whether evidence supports claims regardless of presentation style, (4) whether the essay achieves its communicative purpose. Score 1-9 on the IELTS Coherence and Cohesion band descriptors.

Meta-adjudicator (introduced in Part B)

You are a meta-adjudicator for an essay-scoring panel. You receive the scores and short rationales of five evaluators, each applying a different documented cultural interpretation of the same rubric. Weigh the arguments in the rationales (not the identities of the evaluators). Discount a rationale that misidentifies the essay's structure or contradicts the essay's text. You are not told anything about the writer. Respond with JSON: `{"score": <band 1-9>, "confidence": <0-1>, "rationale": "<3-4 sentences>"}`.

Appendix B

Reproducibility

All experiments are reproducible from the public code repository (`github.com/mikenk2010/vs4139514-Fir`
which contains the pipeline, prompts, configurations, analysis code, and essay-identifier manifests. Corpus texts are not distributed, per the ICNALE licence; they are obtainable free of charge through the ICNALE registration process.

Pipeline

Experiments are driven by a single configuration file (`configs/experiment.yaml`) specifying models, conditions, cohorts, and the sampling seed (4139514). Every scoring call uses temperature 0.0 with structured JSON output, is cached on disk keyed by (model, agent, essay, prompt), and is logged as one JSON line; runs are resumable. The sampling manifest (essay identifiers only) is generated by `src/sample_icnale.py`; the GRA validation manifest and the human-dispersion ground truth are generated by `src/make_gra_manifest.py` from the archive's rating workbook.

Analysis

All results tables and figures in Chapter 4 are generated by `src/make_tables.py` and `src/sweep_weights.py` directly from the experiment logs; no statistic is computed or transcribed by hand. Quadratic Weighted Kappa, bootstrap confidence intervals (10,000 resamples, fixed seed), BDS, and ROC calibration are implemented in `src/metrics.py` with unit tests.

Model versions

The exact model identifiers requested through each provider's API, as pinned in `configs/experiment.yaml` and recorded in every log entry, were: `gpt-4o` (OpenAI); `claude-sonnet-4-6` (Anthropic); `qwen3:14b` (Alibaba, open weights, executed locally via Ollama); and `gemini-3.1-pro-preview` (Google). One model change occurred mid-study: the study began with `gemini-2.5-pro`, which the provider withdrew from new API access partway through data collection. The 249 calls completed against it were archived and excluded from all analyses (mixing scores from two model versions within one experimental cell would confound the comparison), and the full Gemini condition was rerun from scratch on `gemini-3.1-pro-preview`. The archived log is retained in the repository for audit. All scoring calls used temperature 0.0; the sampling manifest was frozen with seed 4139514 before any scoring began.

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